

Examiner Marking Guidelines & Scoring Rubric

1. General Principles & Examination Rules

- **Mathematical Precision:** All calculations must be step-by-step, showing explicit substitution into correct mathematical formulas. Final answers without clear supporting working receive zero marks for accuracy.
- **Carry-Through Errors (Follow-Through Marking):** If a candidate makes an arithmetic error early in a multi-step calculation but uses the correct method and formula subsequently, award partial credit for the remaining correct working using the carried-forward value.
- **Units of Measurement:** Final values must include proper monetary units (UGX). Omission of required units results in a deduction under Accuracy/Coherence.
- **All the 2 items are compulsory, don't award for unanswered items.**
- **Be consistent throughout for all students strictly applying the same rules throughout, if you had marked incorrect for a particular answer, mark that incorrect throughout and don't change at any point.**
- **The RACE Competency Framework:** Assess every part across all four criteria:
 - **R – Relevance:** Directly applies the requested mathematical formulas and financial concepts to the scenario context.
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 - **A – Accuracy:** Correct substitution, precise arithmetic, exact formulas, and correct final numerical values.
 - **C – Coherence:** Clear, logical, step-by-step layout, proper mathematical notation, and clean structural flow.
 - **E – Excellence:** Demonstrates thorough interpretation of results, comprehensive analytical reporting, and complete execution of all criteria.

. Item Marking Breakdown: Item 1 (Total: 20 Marks)

- **Part (a) [10 Marks]:**
 - **Total Revenue Calculation (8 Marks):**
 - Phase 1 bag count & revenue calculation (2 Marks)
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 - Phase 2 bag count & revenue calculation (3 Marks)
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 - Phase 3 bag count & revenue calculation (2 Marks)
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 - Total Revenue summation (1 Mark)

- **Net Profit Calculation (2 Marks):** Correct summation of total expenses and subtraction from Total Revenue.
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- **Part (b) [10 Marks]:**
 - **Profit Distribution (4 Marks):** Division of net profit among 40 members with a clear reporting layout.
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 - **Comparative Analysis of Average Selling Price (6 Marks):**
 - Calculation of overall weighted average price per bag in phased sales (2 Marks)
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 - Calculation of total revenue if sold at a flat rate of UGX 85,000 per bag (2 Marks)
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 - Comparison and analytical interpretation of phased selling vs. flat rate (2 Marks)
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Detailed Model Answers & Exemplars

ITEM 1 MODEL ANSWERS

Part (a): Calculation of Total Revenue and Net Profit (10 Marks)

Correct Answer (Full Marks Exemplar)

Total Harvest: 1,200 bags

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1. Phase 1 Sales:

- Quantity sold = 31 bags

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- Price per bag = UGX 80,000

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- **Phase 1 Revenue:** $31 \times 80,000 = \text{UGX } 2,480,000$

2. Phase 2 Sales:

- Remaining bags after Phase 1 = $1,200 - 31 = 1,169$ bags

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- Quantity sold = 60% of 1,169 = $10060 \times 1,169 = 701.4$ bags

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- Price per bag = UGX 100,000

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- **Phase 2 Revenue:** $701.4 \times 100,000 = \text{UGX } 70,140,000$

3. Phase 3 Sales:

- Remaining bags after Phase 2 = $1,169 - 701.4 = 467.6$ bags

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- Price per bag = UGX 70,000

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- **Phase 3 Revenue:** $467.6 \times 70,000 = \text{UGX } 32,732,000$

4. Total Revenue:

Total Revenue = $2,480,000 + 70,140,000 + 32,732,000 = \text{UGX } 105,352,000$

5. Total Expenses & Net Profit:

- Transport cost = UGX 2,500,000

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- Storage fees = UGX 1,500,000

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- Total Costs = $2,500,000 + 1,500,000 = \text{UGX } 4,000,000$

- **Net Profit:**

Net Profit = Total Revenue - Total Costs = $105,352,000 - 4,000,000 = \text{UGX } 101,352,000$

Incorrect Answer (Zero/Low Marks Exemplar)

$$1,200 - 31 = 1,169.$$

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$$31 \times 80,000 = 2,480,000.$$

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$$1,169 \times 100,000 = 116,900,000. \text{ Total Revenue} = 119,380,000. \text{ Net Profit} = 119,380,000 - 2,500,000 = 116,880,000.$$

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Examiner Note: Fails to calculate the 60% portion for Phase 2, ignores Phase 3 entirely, and leaves out storage expenses. Zero marks for accuracy and method.
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Part (b): Analytical Report for Chairperson (10 Marks)

Correct Answer (Full Marks Exemplar)

FINANCIAL ANALYTICAL REPORT TO: Chairperson, Luweero Farmers' Group
SUBJECT: Profit Distribution and Selling Strategy Analysis
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1. Profit Distribution Among Members (4 Marks):

- Total Net Profit = UGX 101,352,000
- Total Number of Members = 40

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- **Share per Member:**

Individual Share = $40 \text{UGX } 101,352,000 = \text{UGX } 2,533,800$

Each of the 40 members will receive **UGX 2,533,800** as their equal share of net profits.

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2. Comparative Impact of Phased Selling vs. Flat Rate Selling (6 Marks):

- **Actual Effective Average Price per Bag:**

Average Price = $\frac{\text{Total Bags}}{\text{Total Revenue}} = \frac{1,200 \text{UGX } 105,352,000}{\approx \text{UGX } 87,793.33 \text{ per bag}}$

- **Alternative Strategy (Selling all at flat rate of UGX 85,000 per bag):**

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Revenue at Flat Rate = $1,200 \times 85,000 = \text{UGX } 102,000,000$

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- **Comparative Financial Gain:**

Extra Revenue Earned = $105,352,000 - 102,000,000 = \text{UGX } 3,352,000$

Conclusion & Insights: Selling in phases was significantly more profitable than selling all bags at once. Capitalizing on high market demand in Phase 2 at UGX 100,000 per bag raised the group's effective average selling price to **UGX 87,793.33** per bag (compared to the flat rate of UGX 85,000). This strategy generated an additional **UGX 3,352,000** in total revenue for the group.

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Incorrect Answer (Zero/Low Marks Exemplar)

Every member gets 100,000 shillings. Selling all at once for 85,000 is better because Phase 1 was 80,000 and Phase 3 was 70,000 which are less than 85,000.

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Examiner Note: Provides incorrect profit shares without mathematical backing and draws a flawed conclusion by ignoring the large volume sold at UGX 100,000 in Phase 2. Award 0 marks.

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Item 2: Coordinate Geometry, Area, and Budgeting (Total: 20 Marks)

- **Part (a) [10 Marks]:**

- **Perimeter Calculation (5 Marks):**

- Correct application of distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ for sides AB, BC, and AC (3 Marks)
 - Summation of side lengths to find total perimeter (2 Marks)

- **Area Calculation (5 Marks):** Correct application of Shoelace formula or Base/Height method to find exact area of triangle ABC in m^2 .

- **Part (b) [10 Marks]:**

- **Fencing Rolls Needed (3 Marks):** Calculation of rolls needed with explicit justification for rounding up to the nearest whole integer.
 - **Fertilizer Bags Needed (3 Marks):** Calculation of bags needed with explicit justification for rounding up to the nearest whole integer.
 - **Total Material Cost (4 Marks):** Correct multiplication of whole units by unit prices and summation of total budget cost.

Item 2: Coordinate geometry

Part (a) Model answer

Points are:

- A(2,3)
- B(10,3)
- C(6,11)

Step 1: Find the side lengths.

Distance AB:

$$AB=(10-2)^2+(3-3)^2=8^2+0^2=8$$

Distance BC:

$$BC=(6-10)^2+(11-3)^2=(-4)^2+8^2=16+64=80=45$$

Distance CA:

$$CA=(2-6)^2+(3-11)^2=(-4)^2+(-8)^2=16+64=80=45$$

Step 2: Find the perimeter.

$$P=AB+BC+CA=8+45+45$$

$$P=8+85$$

Approximate value:

$$mP \approx 8+8(2.236)=8+17.888=25.888 \text{ m}$$

So, **perimeter** = $8+85 \text{ m} \approx 25.89 \text{ m}$.

Step 3: Find the area.

Since AB is horizontal, use base $AB=8 \text{ m}$ and height from C to line AB.

Line AB is at $y=3$, so the height is:

$$11-3=8 \text{ m}$$

Area:

$$A=\frac{1}{2} \times 8 \times 8=32$$

So, **area** = 32 m^2 .

Part (b) Model answer

Fencing rolls needed

Perimeter $\approx 25.89 \text{ m}$.

One roll covers 10 m , so:

$$10 \div 25.89 = 2.589$$

Since partial rolls cannot be bought, round up:

rolls 3 rolls

Cost of fencing:

$$3 \times 150,000 = 450,000$$

Fertilizer bags needed

Area $\text{m}^2 = 32 \text{ m}^2$.

One bag covers 12 m^2 , so:

$$12 \div 32 = 2.666\dots$$

Round up:

bags 3 bags

Cost of fertilizer:

$$3 \times 25,000 = 75,000$$

Total cost

$$450,000 + 75,000 = 525,000$$

*So, **total cost** = UGX 525,000.*

Why rounding up is necessary

In real life, fencing wire and fertilizer are sold only in whole units. If the required amount is 2.589 rolls, buying 2 rolls would not be enough, so 3 rolls must be purchased. The same applies to fertilizer: 2.666 bags still requires 3 ba